Progress Report

Project Overview

The purpose of this project is to explore the relationship between AI usage and career anxiety among university students using the AI Dependency, Career Anxiety, and Student Burnout dataset. The client is Face The Future Counselling (FTF) which is a counselling startup that wants to better understand how the increasing use of artificial intelligence is affecting students' mental health, career readiness, and professional development across different fields of study. The client specifically asks for specific harmful patterns in mental health that arise in different areas of study in correlation to AI use. They also want to know if there are patterns in student habits that help lower career anxiety and support higher career readiness - which may differ depending on whether a student heavily uses AI or not. Lastly, the client also wants to know which areas of study involve high-effort students who use a lot of AI with the highest amount of career stress.

Progress Completed

Right now the dataset has been successfully imported into Jupyter Notebook, and the data cleaning phase has been completed. I started with an initial exploration of the dataset to understand its size, variable types, description, and overall quality. This helped me identify several variables containing missing values and gave me a clear overview of the data before moving into the analysis portion.

After taking a closer look at the missing values, I discovered that most variables had only a small number of missing observations. However, the primary_ai_tools_used variable contained approximately 21% missing values. Instead of removing a significant portion of records, I utilized different imputation methods based on the type of variable and the amount of missing data. Numerical variables with relatively few missing values were filled using the median because it is less affected by skewed distributions and extreme values. The field seeks_career_counseling was imputed using the mode to preserve the most common response. Meanwhile, the primary_ai_tools_used field missing values were retained as an "Unknown" category which helped preserve more than 3,000 student records to remain in the dataset.

After addressing the missing values, I checked the dataset for duplicate records to ensure that each record represented a unique student. An initial outlier assessment was also carried out using boxplots for the continuous variables. Although some observations fell outside, these values seem to represent realistic student behaviours rather than data entry errors. For instance, higher levels of daily AI usage, study hours, and weekly job applications are all reasonable among university students and may be important for identifying students who are highly motivated but also experiencing greater stress. As a result, no outliers were removed from the dataset.

Currently, I have also created plots to observe study area distribution, ai usage distribution, career anxiety distribution, and a correlation matrix. Although these plots have been made, I am yet to further analyze these plots to derive possible patterns and conclusions.

Next Steps

Moving forth, a greater exploration of these relationships will be conducted and I will focus on answering the client's research questions through even more detailed analysis and visualizations. This includes examining AI usage across different areas of study, investigating relationships between AI dependency, burnout, placement anxiety, interview anxiety, and career readiness, and identifying study habits that are associated with lower career anxiety and stronger career readiness.